From Random Walks to Sequential Testing

The Complete Statistical Journey in 14 Acts

The story. Sequential testing enables you to peek at test results as often as you like and the false positive guarantee still holds. Getting there has required several developments in statistics. This webpage takes you through that journey, one step at a time: beginning with coin flips and random walks, proceeding with gambling examples in 1939 and testing ammunition in WWII, all the way up to modern confidence sequences.

Format. Fourteen interactive acts. Each act introduces one concept and builds on the previous ones.

Target audience. Data scientists and analysts familiar with A/B testing, hypothesis testing, and confidence intervals.

Act 1
The Peeking Problem

Act 2
Random Walk

Act 3
Martingale

Act 4
Likelihood Ratio

Act 5
LR is a Martingale

Act 6
Ville's Inequality

Act 7
Wald's SPRT

Act 8
Mixture / mSPRT

Act 9
Confidence Sequences

Act 10
Problem Eppo Solves

Act 11
Eppo Pipeline

Act 12
Variance Reduction

Act 13
Sequential CI

Act 14
DIY Alternatives

Act 1 — Why Should You Care? The Peeking Problem

The Motivation

Consider a standard A/B test comparing a control experience against a new variant. After a week of data collection, the dashboard reports a desired effect with $pext{-}value = 0.03$. At $alpha = 0.05$, this is statistically significant, so the variant is declared a winner.

But here is the problem: if you monitored the test daily before that decision, the actual false positive rate may be far higher than 5%. A substantial fraction of "significant" results under this practice are false positives — driven by random fluctuation rather than a genuine treatment effect.

Simulation

Setup: 1,000 A/B experiments simulated under the null hypothesis (zero treatment effect). Both groups are drawn from identical distributions. A standard t-test is recomputed after each new observation.

The cumulative fraction of experiments that show $p < 0.05$ at some point during monitoring climbs well above 5%.

Peeking Simulation

Run 500 A/A experiments (no real effect) and count how often you get a false positive under different peeking schedules.

Run simulation

Why Does This Happen?

When you run a standard hypothesis test with $alpha = 0.05$, the 5% false positive guarantee holds only if you commit to a fixed sample size and analyse the result once.

If you monitor the experiment and repeatedly check for significance, that guarantee no longer holds. Each interim analysis gives random fluctuations another opportunity to cross the significance threshold.

The tests across interim looks are not independent, but the overall effect is clear: the more often you peek — and especially if you stop as soon as $p < 0.05$— the higher the realised false positive rate.

The rest of this presentation develops a testing framework that does not suffer from this problem: one where the Type I error guarantee holds regardless of the number or timing of analyses.

That framework is called **anytime-valid inference**.

How Bad Is It?

Checking schedule	Actual false positive rate
Once at the end (as designed)	$\sim 5\%$
Daily for 1 week	$\sim 13\%$
Daily for 2 weeks	$\sim 19\%$

Checking schedule	Actual false positive rate
Daily for 4 weeks	$\sim 25\%$
Continuously (every observation)	Can exceed 30%

Key Takeaway

The peeking problem: Repeated interim analyses of a standard hypothesis test inflate the Type I error rate well beyond its nominal level. Valid continuous monitoring requires a fundamentally different testing framework — one whose error guarantees hold regardless of the stopping rule.

Act 2 — The Random Walk

Simulation

A symmetric random walk starting at the origin. At each step, $+1$ (heads) or -1 (tails) with equal probability. The cumulative sum S_n is plotted against the step number.

Random Walk Simulation

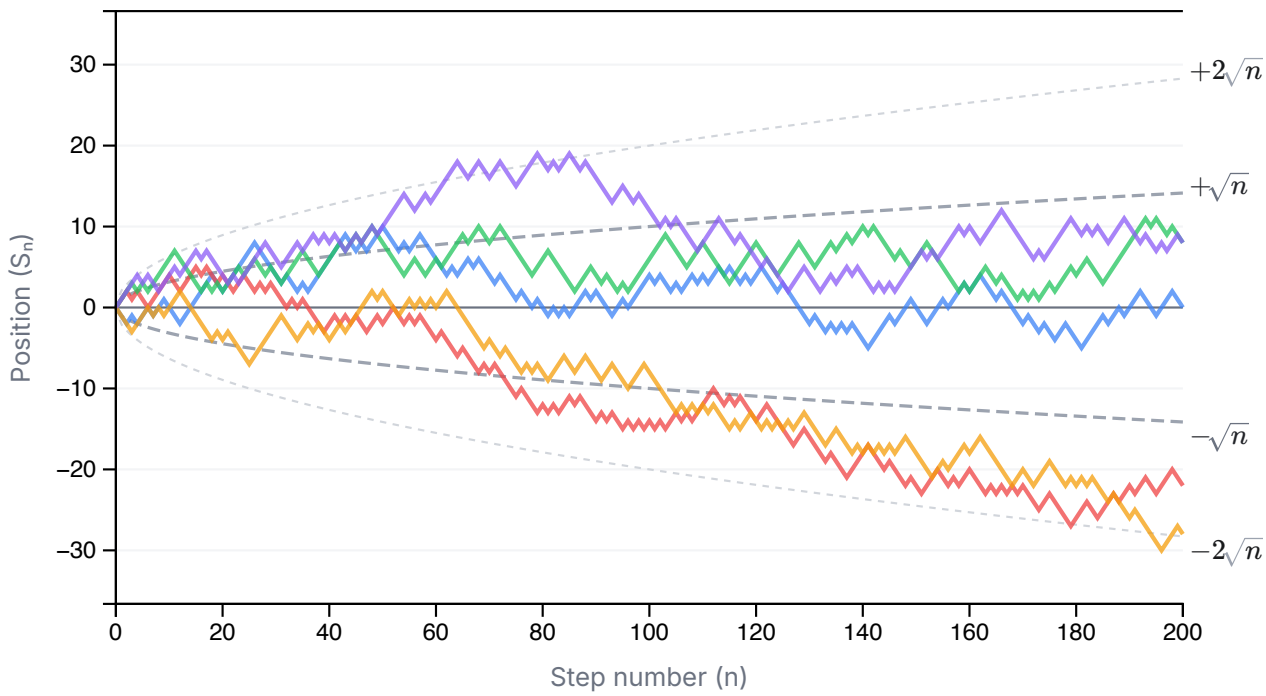
Watch coin-flip random walks fan out over time. Dashed lines show the ±√n and ±2√n envelopes.

Run simulation

Steps: 200

P(heads): 0.50

Paths: 5



Simulation takeaway: Each path is a cumulative sum of ±1 coin flips. With a fair coin (p = 0.5), the expected sum of flips is zero. About 68% of the time the walk stays within $\pm\sqrt{n}$ of zero, and about 95% within $\pm 2\sqrt{n}$.

A random walk is the simplest stochastic process: independent, identically distributed increments summed cumulatively. The walk has zero drift (the expected position at any step is the current sum) but increasing dispersion.

Individual paths can wander far from zero, but the distribution of S_n across many realisations is centred at zero with standard deviation $sqrtn$. This spreading has a precise characterisation via the Central Limit Theorem.

Mathematical Formulation

Setting up the notation

Let X_i represent the outcome of the ith coin flip:

$$X_i = \begin{cases} +1 & \text{if heads (step forward)} \\ -1 & \text{if tails (step backward)} \end{cases}$$

The probability of heads is written p. For a fair coin, $p = 0.5$.

Position after n steps

$$S_n = \sum_{i=1}^{n} X_i = X_1 + X_2 + \cdots + X_n$$

In words: S sub n equals the sum of all X_i, starting from $i = 1$ up to $i = n$.

Example.
First 5 flips: H, T, H, H, T, so $X_1 = +1,\ X_2 = -1,\ X_3 = +1,\ X_4 = +1,\ X_5 = -1.$

$$S_1 = 1, \quad S_2 = 0, \quad S_3 = 1, \quad S_4 = 2, \quad S_5 = 1$$

The average outcome (expected value)

When $p = 0.5$, each flip has expected value zero:

$$\mathbb{E}[X_i] = \underbrace{0.5}_{P(\text{heads})} \times (+1) + \underbrace{0.5}_{P(\text{tails})} \times (-1) = 0$$

Because each step averages to zero:

$$\mathbb{E}[S_n] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \cdots + \mathbb{E}[X_n] = 0$$

On average, across many trials, you end up back where you started.

How spread out the outcomes are — variance & standard deviation

Variance measures the average squared distance from the mean. For a single fair co

$$\mathrm{Var}(X_i) = \mathbb{E}[(X_i - 0)^2] = 0.5 \times (+1)^2 + 0.5 \times (-1)^2 = 1$$

Because flips are independent, the variance of the sum is the sum of the variances:

$$\mathrm{Var}(S_n) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2) + \cdots + \mathrm{Var}(X_n) = \underbrace{1 + 1 + \cdots + 1}_{n \text{ terms}} = n$$

The **standard deviation**is the square root of the variance — same units as the position:

$$\mathrm{SD}(S_n) = \sqrt{\mathrm{Var}(S_n)} = \sqrt{n}$$

Before we begin flipping the coin, the expected position is zero: $\mathbb{E}[S_n] = 0$. But positions spread out over time. If we flip the coin a large number of times, about 68% of walks would end up within $\pm\sqrt{n}$ of zero (one standard deviation), and about 95% within $\pm 2\sqrt{n}$ (two standard deviations). More generally, the distribution of the sum of walks would be approximately normal. For example, after 100 steps, 68% of walks are within ±10 and 95% within ±20. The spread grows with the *square root* of the number of steps, not the number itself.

What if the coin is not fair?

When $p \neq 0.5$, each step has a non-zero average:

$$\mathbb{E}[X_i] = p \times (+1) + (1 - p) \times (-1) = 2p - 1$$

For $p = 0.6$: $\mathbb{E}[X_i] = 0.2$. After n steps:

$$\mathbb{E}[S_n] = n(2p - 1)$$

This is a systematic *drift* — the random walk now has a trend.

Key Takeaway

Random fluctuations can cause the sum of many coin flips to be far from zero, even though the expected value of each coin flip is zero.

Act 3 — The Gambling Game and the Martingale

Imagine flipping a fair coin, winning $+1$ for heads and losing -1 for tails. Can you walk away a winner by picking the perfect moment to stop?

Theorem (Doob's Optional Stopping Theorem)

Let M_n be a martingale and au a **stopping time**— a rule for when to quit that uses only information accumulated so far (no looking into the future). Then, under reasonable conditions:

$$EE[M_a u] = EE[M_0]$$

For a gambler starting with $M_0 = 0$: expected winnings at stopping = 0, always, for any strategy. **You cannot beat a fair game by choosing when to quit.**

What is a stopping time?

τ is a decision rule of the form: “Stop after step n if [condition based only on what happened so far].”

Rule	Valid?
Stop after exactly 100 flips	✓
Stop the first time I am ahead by €10	✓
Stop when I have lost €50	✓
Stop after 1000 flips *or* when ahead by €5, whichever first	✓
Stop right before the next loss	✗ (requires the future!)

The “reasonable conditions” for Doob's theorem

Doob's theorem requires at least one of:

1. $\tau \leq N$ for some fixed N (bounded stopping time), **or**
2. $|M_n| \leq C$ for some fixed C (bounded martingale), **or**
3. $\mathbb{E}[\tau] < \infty$ and certain integrability conditions (finite expected stopping time).

When *none* hold — as with the doubling strategy — the guarantee breaks down.

Why the doubling strategy fails

The doubling strategy guarantees eventual profit *only if* you have infinite money and infinite time. In practice:

- Finite budget ⇒ real probability of ruin before recovery.
- Expected time to recovery is infinite: $\mathbb{E}[\tau] = \infty$.
- Doob's theorem gives $\mathbb{E}[M_\tau] \leq 0$, not $= 0$.

Why this matters for what is coming

The martingale concept reappears in every subsequent act. The likelihood ratio (Act 5) is a martingale. The mixture likelihood ratio (Act 8) is a martingale. The reason sequential tests control false positives — even with peeking — is that they exploit the same structure: **a fair game cannot be beaten by choosing when to stop.**

Key Takeaway

A fair game cannot be beaten by choosing when to stop.

Historical Note

Joseph L. Doob formalised martingale theory in his 1953 monograph *Stochastic Processes*. The name "martingale" comes from a French term for a type of betting strategy — the very doubling strategy we just debunked.

Act 4 — Hypothesis Testing: Telling Two Coins Apart

You're handed a coin. Is it fair ($p = 0.5$) or biased ($p = 0.5 + delta$)? With just a few flips, the difference is invisible. With hundreds, the truth emerges.

Null hypothesis H_0: The coin is fair.
Alternative hypothesis H_1: The coin is biased.

Type I error (false positive): You call the coin biased when it's actually fair.
Type II error (false negative): You call the coin fair when it's actually biased.

Standard thresholds: $alpha = 0.05$ (max false positive rate), eta = 0.20 (max false negative rate). **Power** = 1 - eta = 0.80.

The more data you collect, the easier it is to tell the difference. But with small bias, even long sequences can be ambiguous.

Simulation

Flip a coin and try to decide: is it fair or biased? The simulation shows how hard it is to distinguish the two, especially when the bias is small. Over short runs, the sequences look nearly identical.

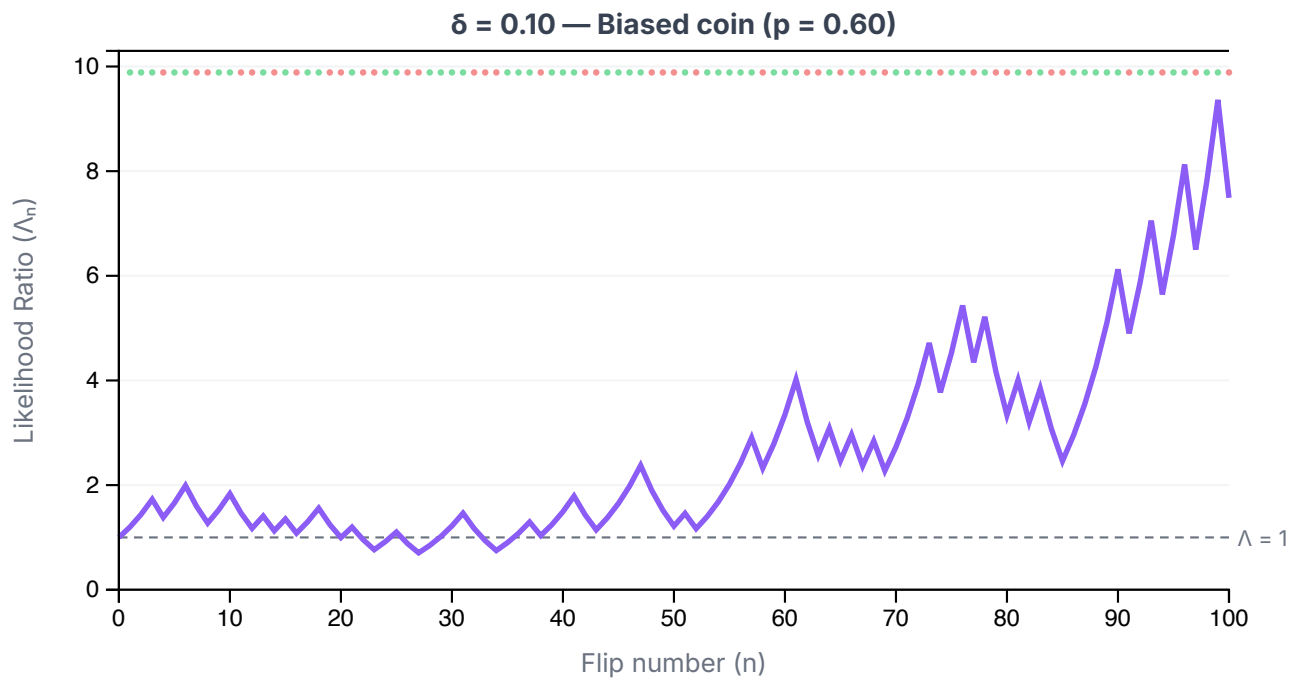
Likelihood Ratio Accumulation

Watch the LR grow as evidence accumulates flip by flip. Green dots = heads, red dots = tails.

Run simulation

Bias δ: 0.10 → coin p = 0.60

Flips: 100



7.49	60/100	Favours H_1
Final Λ_n	Heads	Moderate evidence

Simulation takeaway: Even with a real bias, it takes a lot of data to reliably tell the difference. Short sequences are noisy; long sequences reveal the truth.

Mathematical Formulation

The likelihood

Suppose you flip the coin n times and see k heads.

The likelihood if the coin is fair (H_0):

$$mathcalL_0 = (0.5)^k imes (0.5)^{n-k} = (0.5)^n$$

The likelihood if the coin is biased (H_1):

$$mathcalL_1 = (0.5 + delta)^k imes (0.5 - delta)^{n-k}$$

The likelihood ratio

Lambda_n = rac{mathcal{L}_1}{mathcal{L}_0} = rac{(0.5+delta)^k ;(0.5-delta)^{n-k}}{(0.5)^n}

The likelihood ratio tells you how much more likely the data are if the coin is biased versus fair.

Example.
$delta = 0.1$ (biased coin has $p = 0.6$); $n = 10$ flips, $k = 7$ heads.

egin{aligned} mathcal{L}_1 &= (0.6)^7 imes (0.4)^3 approx 0.001792 \ mathcal{L}_0 &= (0.5)^{10} approx 0.000977 \ Lambda_{10} &= rac{0.001792}{0.000977} approx 1.83 end{aligned}

The data is $approx 1.83$ times more likely if the coin is biased.

Interpreting Λ_n

Λ_n	**Interpretation**
$= 1$	Data equally consistent with both hypotheses
$= 10$	10× more likely under H_1 — moderate evidence
$= 100$	100× more likely under H_1 — strong evidence

Λ_n	Interpretation
$= 0.1$	10× more likely under H_0 — evidence for fairness

Incremental updating

Λ_n can be computed one flip at a time:

$$\Lambda_n = \Lambda_{n-1} \times \frac{f_1(x_n)}{f_0(x_n)}$$

where $f_j(x_n)$ is the probability of the nth flip's outcome under H_j.

- Heads: multiply by $\frac{0.5+\delta}{0.5}$.
- Tails: multiply by $\frac{0.5-\delta}{0.5}$.

This incremental structure is what makes *sequential* testing possible.

Starting value

Before any data: $\Lambda_0 = 1$. Neither hypothesis is favoured. This will be important for Ville's inequality (which requires the starting value to equal 1).

Key Takeaway

The likelihood ratio quantifies how many times more probable the observed data are under H_1 relative to H_0.

Act 5 — The Likelihood Ratio Is a Martingale

Under H_0 (the coin is fair), the likelihood ratio $Lambda_n$ bounces up and down, but on average stays at 1. No matter how you peek or stop, you can't systematically make it large. This is the martingale property in action.

Key point: The likelihood ratio is just like the gambler's winnings in Act 3. No clever strategy can beat the system if the coin is fair.

Simulation

The plot shows 100 paths of $Lambda_n$ over time. When the coin is fair (H_0), the paths wander with no trend. When the coin is biased (H_1), the paths drift upward.

LR Paths: H_0 vs H_1

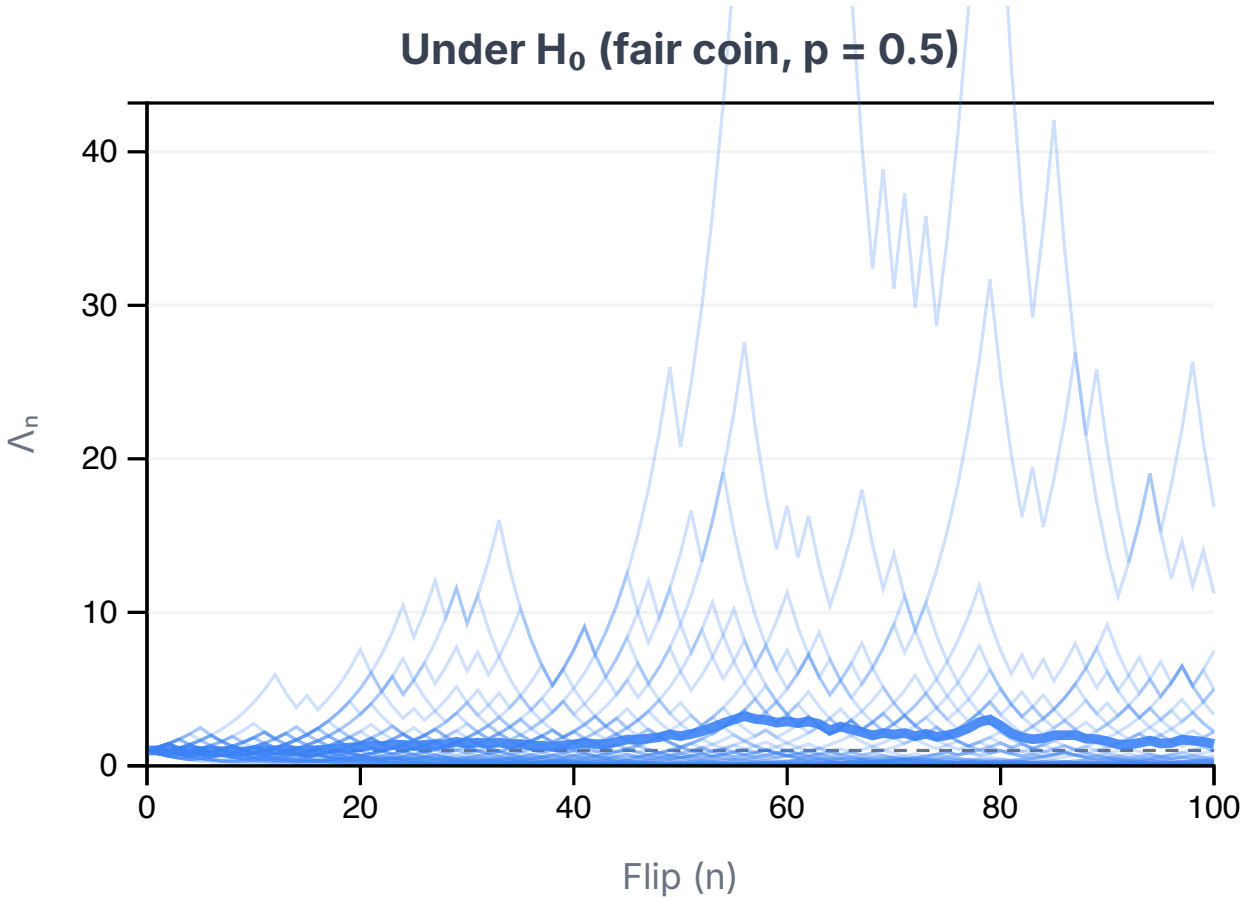
50 likelihood ratio paths. Under H_0 they wander (martingale); under H_1 they drift up. Bold line = average path.

Run simulation

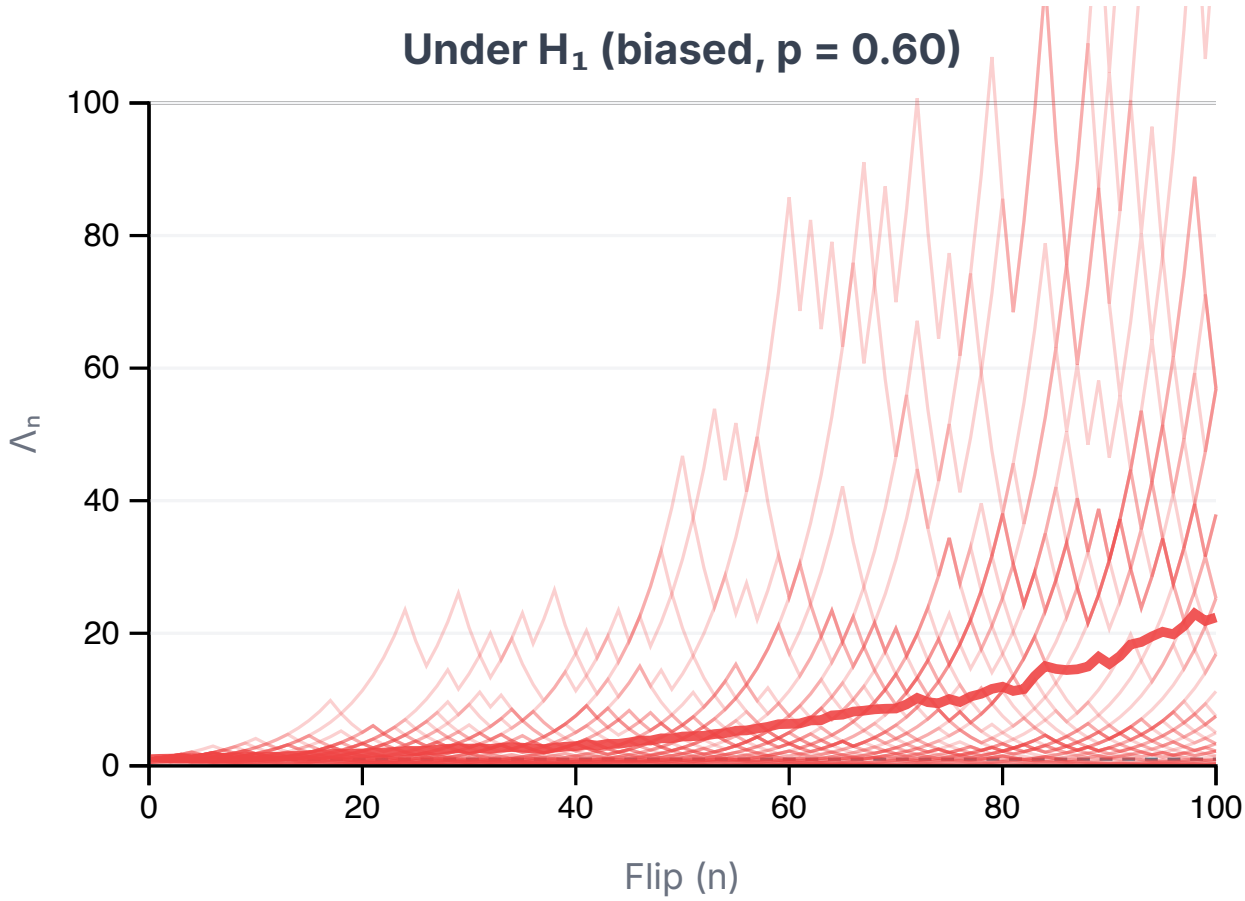
δ: 0.10

Flips: 100

Paths: 50



Mean final Λ: 1.42 (should ≈ 1)



Mean final Λ: 33.18 (drifts upward)

Simulation takeaway: If the coin is fair, the likelihood ratio stays around 1, no matter how you peek. If the coin is biased, it grows. This is the core of sequential testing.

Mathematical Formulation

Proof that Λ_n is a martingale under H_0

We must show: $\mathbb{E}[\Lambda_n \mid \Lambda_0, \ldots, \Lambda_{n-1}] = \Lambda_{n-1}$.

From Act 4, $\Lambda_n = \Lambda_{n-1} \times \frac{f_1(x_n)}{f_0(x_n)}$. Therefore:

$$\begin{aligned}\mathbb{E}[\Lambda_n \mid \text{past}] &= \mathbb{E}\left[\Lambda_{n-1} \cdot \frac{f_1(x_n)}{f_0(x_n)} \;\middle|\; \text{past}\right] \\ &= \Lambda_{n-1} \cdot \mathbb{E}\left[\frac{f_1(x_n)}{f_0(x_n)}\right] \qquad (\Lambda_{n-1} \text{ is known})\end{aligned}$$

Now the key step. Under H_0, x_n is drawn from f_0. So:

$$
\begin{aligned}
\mathbb{E}\left[\frac{f_1(x_n)}{f_0(x_n)}\right] &= \sum_{\text{all outcomes } x} f_0(x) \cdot \frac{f_1(x)}{f_0(x)} \\
&= \sum_x \cancel{f_0(x)} \cdot \frac{f_1(x)}{\cancel{f_0(x)}} \quad \textbf{(the cancellation!)} \\
&= \sum_x f_1(x) \\
&= 1 \quad (f_1 \text{ is a probability distribution})
\end{aligned}
$$

Therefore:

$$\mathbb{E}[\Lambda_n \mid \text{past}] = \Lambda_{n-1} \times 1 = \Lambda_{n-1} \quad \checkmark$$

The cancellation with actual numbers

For $\delta = 0.1$ (biased coin $p = 0.6$):

Outcome	$f_0(x)$	$f_1(x)/f_0(x)$	$f_0(x) \times [f_1(x)/f_0(x)]$
Heads	0.5	$0.6/0.5 = 1.2$	$0.5 \times 1.2 = 0.6$
Tails	0.5	$0.4/0.5 = 0.8$	$0.5 \times 0.8 = 0.4$
Total:			**1.0 ✓**

When we compute the expected ratio under H_0, the H_0 probabilities cancel out, leaving the sum of the H_1 probabilities — which equals 1 because f_1 is a valid probability distribution.

Two additional properties

1. Λ_n **is non-negative:** It is a product of non-negative ratios (probabilities divided by probabilities), so $\Lambda_n \geq 0$.
2. $\Lambda_0 = 1$**:** Before any data, neither hypothesis is favoured.

So Λ_n is a **non-negative martingale starting at 1**. Remember these properties — they

exactly what Ville's inequality needs.

Under H_1, Λ_n drifts upward

When the coin really is biased:

$$\mathbb{E}\left[\frac{f_1(x_n)}{f_0(x_n)}\right]_{H_1} = \sum_x f_1(x) \cdot \frac{f_1(x)}{f_0(x)} > 1$$

So $\Lambda_n$tends to grow — this upward drift is the signal the test detects.

Key Takeaway

Key concepts: the likelihood ratio is a martingale under H_0; the cancellation that makes it work; non-negative martingale; drift under H_1.

Act 6 — Markov, Ville, and the Anytime-Valid Guarantee

This is the pivotal act: why can you peek at your test statistic after every observation, and still control the false positive rate?

Two steps:

1. **Markov's inequality:** gives a bound at a single, fixed time.
2. **Ville's inequality:** extends the bound to all times, no matter how often you peek.

Ville's inequality is the reason sequential testing works. It's the mathematical engine behind the anytime-valid guarantee.

Simulation

Panel 1 — Markov (single time): 10,000 fair-coin paths of $Lambda_n$. Threshold at $Lambda = 1/alpha = 20$. How many are above the threshold at step 200? Fraction $leq5$.

Panel 2 — Ville (any time): Same paths. How many ever cross the threshold at any step? Also $leq5$—but this needs a stronger theorem.

Panel 3 — Comparison: Standard z-score checked at every step. How many ever show significance? Much higher than 5%—the peeking problem.

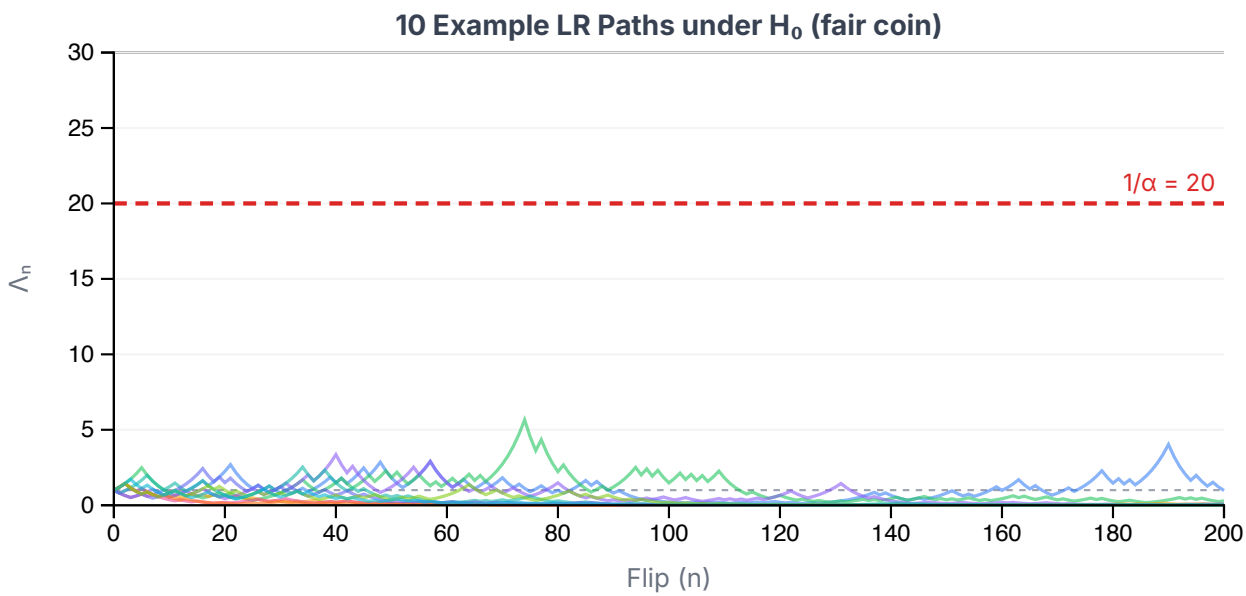
Markov vs Ville vs Peeking z-test

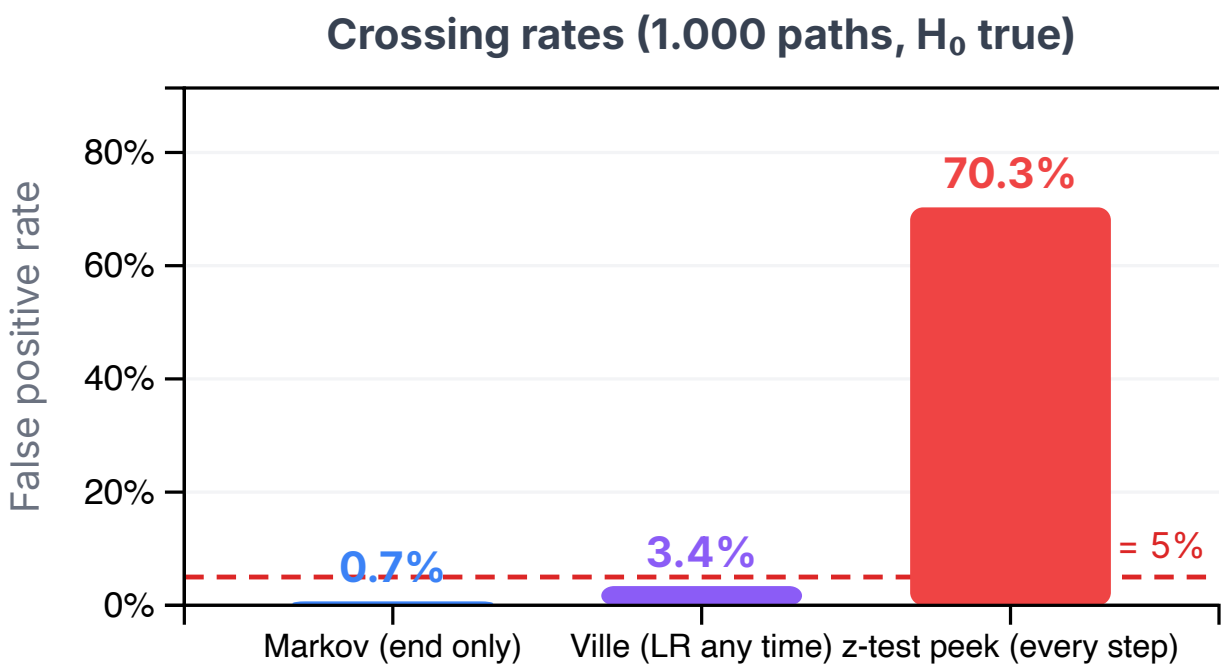
All under H_0 (fair coin). Compare: crossing threshold at one time (Markov), ever crossing (Ville), and repeated z-test significance.

Run simulation

Paths: 1.000

Flips: 200





Key insight: Both Markov (0.7%) and Ville (3.4%) stay below α = 5%, while the standard z-test with peeking (70.3%) far exceeds it. The LR threshold is anytime-valid; the p-value threshold is not.

Simulation takeaway: Markov's inequality controls the false positive rate at a single time. Ville's inequality controls it across all times—even if you peek after every flip. This is the anytime-valid guarantee.

Mathematical Formulation

Step 1: Markov's Inequality

Theorem (Markov's Inequality)

For any non-negative random variable X with finite $\mathbb{E}[X]$:

$$\mathbb{P}(X \geq c) \leq \frac{\mathbb{E}[X]}{c}$$

Why is this true? An intuitive proof.

Suppose $X \geq 0$ always. Let $q = \mathbb{P}(X \geq c)$ be the fraction of outcomes at or above c. Those outcomes contribute *at least* $q \times c$ to the average (each is $\geq c$). The rest contribute ≥ 0 (since $X \geq 0$):

$$\mathbb{E}[X] \geq q \cdot c + (1-q) \cdot 0 = q \cdot c \quad \Longrightarrow \quad q \leq \frac{\mathbb{E}[X]}{c} \quad \checkmark$$

Concrete example. Average income in a town is €50,000.

$$\mathbb{P}(\text{income} \geq €500{,}000) \leq \frac{50{,}000}{500{,}000} = 0.10 = 10\%$$

At most 10% of people can earn €500,000+.

Applying Markov to the likelihood ratio.

Under H_0, Λ_n is a martingale with $\mathbb{E}[\Lambda_n] = \Lambda_0 = 1$ and $\Lambda_n \geq 0$. Markov gives:

$$\mathbb{P}\left(\Lambda_n \geq \frac{1}{\alpha}\right) \leq \frac{\mathbb{E}[\Lambda_n]}{1/\alpha} = 1 \cdot \alpha = \alpha$$

At any single, pre-specified step n, the false positive probability is at most α. Reassuring — but only for *one* moment.

Step 2: Ville's Inequality (1939)

Theorem (Ville's Inequality)

For any **non-negative martingale** $\{M_n\}_{n \geq 0}$ with $M_0 = m_0$:

$$\mathbb{P}\left(\sup_{n \geq 0} M_n \geq c\right) \leq \frac{m_0}{c}$$

where $\sup_{n \geq 0} M_n$ is the highest value M_n *ever* reaches.

Why this is remarkable:

Markov	At any single *fixed* moment, $\mathbb{P}(M_n \geq c)$ is small.
Ville	$\mathbb{P}(M_n$ **ever** $\geq c$, at **any** moment$)$ is *also* small.

The intuition behind Ville.

For a non-negative martingale, reaching a high value "uses up" the available expected value. Once high, it must stay high on average (martingale property), yet it cannot go below zero (non-negativity). These two constraints together limit how often it can ever be high.

Applying Ville to the likelihood ratio

Set $M_n = \Lambda_n$, $m_0 = \Lambda_0 = 1$, $c = 1/\alpha$:

$$\mathbb{P}\left(\Lambda_n \text{ ever reaches } \frac{1}{\alpha} \text{ or higher}\right) \leq \alpha$$

The probability that the likelihood ratio ever crosses our rejection threshold — at any point, across all flips, even if you run forever — is at most α.

This is the anytime-valid guarantee. You can peek after every observation. The false positive rate, across all checks combined, stays $\leq \alpha$.

Connecting back to Act 1

Standard test statistics (like p-values from a t-test) are *not* martingales. Checking repeatedly gives many independent chances to be fooled.

The likelihood ratio *is* a martingale under H_0. Ville's inequality converts this into a practical guarantee: **peeking is safe.**

Historical Note

Jean Ville proved this inequality in his 1939 doctoral thesis, *Étude critique de la notion de collectif*. It predated the formal development of martingale theory by Doob (1953) and was largely overlooked for decades. Robbins (1970)was among the first to recognise its practical importance for sequential testing — a connection we explore in Act 8.

Key Takeaway

Key concepts: Markov's inequality (with proof), Ville's inequality, the anytime-valid guarantee, martingale + non-negative = peek-safe.

Act 7 — SPRT: Wald's Sequential Probability Ratio Test (1945)

Imagine you have a coin, but you don't know if it's fair (H_0) or biased (H_1). You want to decide as quickly as possible, but you can peek after every flip.

Wald's answer: After each observation, compute $Lambda_n$. If it gets high enough, call the coin biased. If it drops low enough, call it fair. Otherwise, keep flipping.

Key point: This is the most efficient way to decide between two hypotheses with error guarantees. On average, it needs far fewer observations than any fixed-sample test.

Simulation

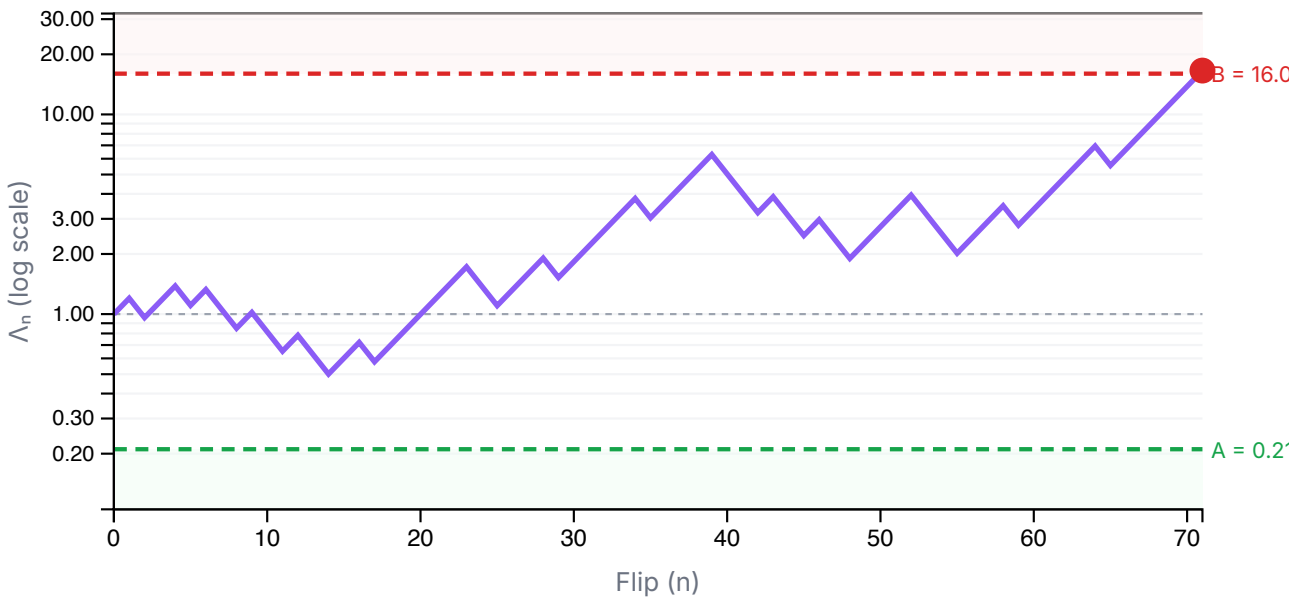
Flip a coin and watch $Lambda_n$ move. If it crosses the upper threshold (B), you call the coin biased. If it falls below the lower threshold (A), you call it fair. Otherwise, keep going.

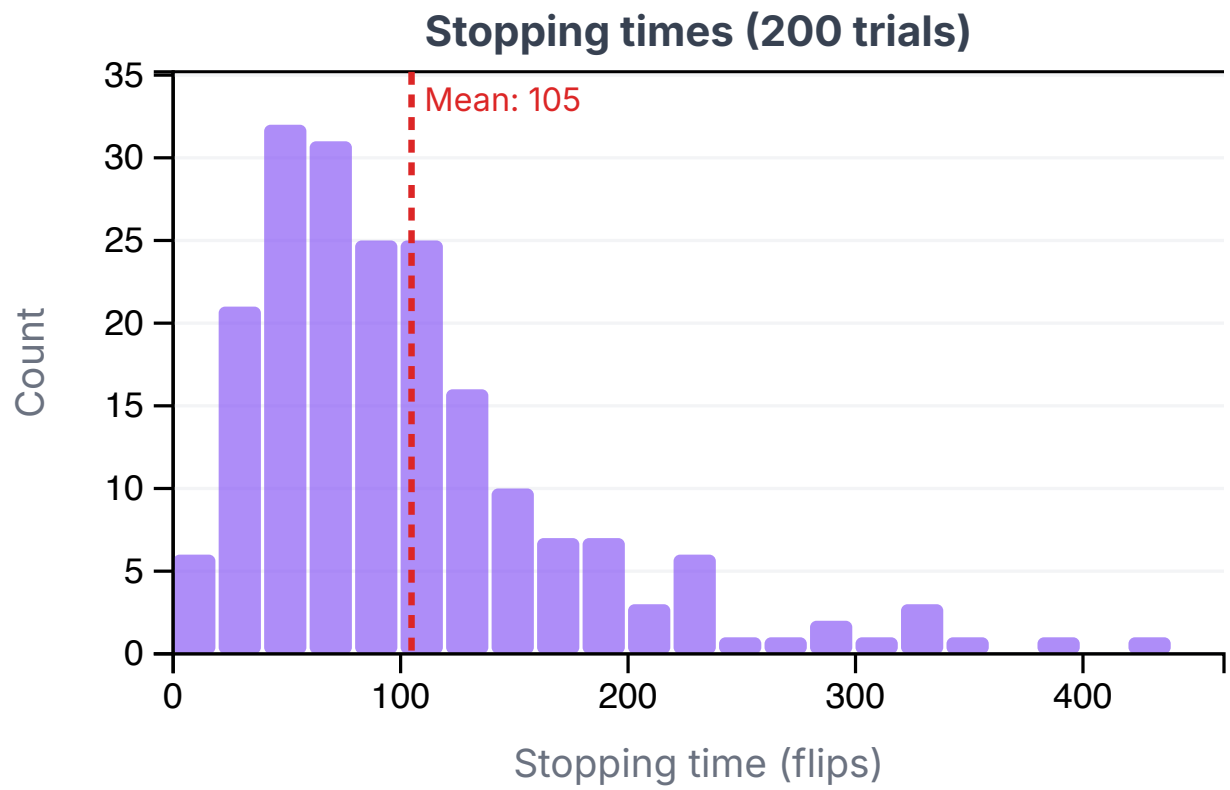
SPRT Decision Boundaries

Watch Λ_n traverse between accept (green) and reject (red) boundaries.

Run simulation

δ: 0.10 α: 0.05 β: 0.20 True coin: Biased (p = 0.60)





159	41	71
Reject H_0 (79.5%)	Accept H_0 (20.5%)	Example stop: Reject H_0 (biased)

Simulation takeaway: The SPRT lets you decide as soon as the evidence is strong enough, saving time and samples. But it requires you to pick the effect size in advance.

Mathematical Formulation

Decision rule

Choose error tolerances α (false positive rate) and β (false negative rate). Compute two thresholds:

$$B = \frac{1-\beta}{\alpha}, \quad A = \frac{\beta}{1-\alpha}$$

Example ($\alpha = 0.05$, $\beta = 0.20$):

$$B = \frac{0.80}{0.05} = 16, \quad A = \frac{0.20}{0.95} \approx 0.211$$

Condition	Decision	Meaning
$\Lambda_n \geq B = 16$	Reject H_0	"Coin is biased"
$\Lambda_n \leq A \approx 0.21$	Accept H_0	"Coin is fair"
$A < \Lambda_n < B$	Keep flipping	"Not enough evidence yet"

Savings over fixed-sample tests

α	β	Saving in expected observations
0.05	0.05	$\approx 53\%$
0.05	0.01	$\approx 47\%$
0.01	0.01	$\approx 63\%$

The SPRT needs, on average, only about **half as many observations** as a traditional fixed-sample test. This was a remarkable result in 1945 and remains one of the main motivations for sequential testing.

The critical weakness

SPRT requires specifying δin advance — the exact alternative hypothesis H_1. If you set $\delta = 0.1$ but the true bias is 0.03, the likelihood ratio is computed against the wrong alternative. The test may take much longer, or still determine the correct answer but far less efficiently.

In A/B testing — where the effect size is typically unknown — this is a serious limitation. The next two acts address this.

Historical Note

Wald conjectured in his 1945 paper that the SPRT is *exactly* optimal. Three years later, Wald & Wolfowitz (1948) proved the general result: among **all** sequential tests with the same or smaller error probabilities (α, β), the SPRT minimises the expected sample size under both H_0 and H_1 simultaneously.

Key Takeaway

Key concepts:SPRT decision rule, Wald's boundary approximations, 47–63% saving over fixed-sample tests, optimality (Wald & Wolfowitz 1948), the critical dependence on a pre-specified effect size.

Act 8 — The Mixture Approach: From Robbins (1970) to the mSPRT

The SPRT is powerful, but it requires you to pick a single effect size in advance. What if you don't know it? Robbins' idea: average the likelihood ratio over a range of plausible effect sizes.

Key point: This "mixture" likelihood ratio is still a martingale under H_0. Ville's inequality still applies. You get the anytime-valid guarantee, but now the test is robust to a range of effect sizes.

The fundamental result

Robbins observed that if Λ_n^δ is the likelihood ratio for a specific effect size δ, then the **mixture likelihood ratio**

$$\Lambda_n^H = \int \Lambda_n^\delta \, dH(\delta)$$

is also a non-negative martingale starting at 1 under H_0, because:

1. Each Λ_n^δ is a non-negative martingale under H_0 (Act 5).
2. A weighted average (integral) of martingales is a martingale (linearity of conditional expectation).

Therefore Ville's inequality gives:

$$\mathbb{P}\left(\Lambda_n^H \text{ ever} \geq \frac{1}{\alpha}\right) \leq \alpha$$

Symbol by symbol:

- Λ_n^δ = the likelihood ratio assuming effect size δ.
- $dH(\delta)$ = how much weight to give effect size δ.
- int = the integral — averages over all possible $delta$.

The result is the expected likelihood ratio under the mixing distribution H, averaging evidence across all plausible effect sizes.

What is the "mixture τ"?

The mixing distribution H is your *prior belief* about how big the true effect could plausibly be. A common choice is $H = \mathcal{N}(0, \tau^2\sigma^2)$— a Normal centred at zero with spread controlled by the parameter τ.

- **Small** τ— you assume the effect (if any) is small. The test is most powerful for small effects, less powerful for large ones.
- **Large** τ— you allow for potentially large effects. The test gains power against big effects but pays for it with wider confidence intervals.

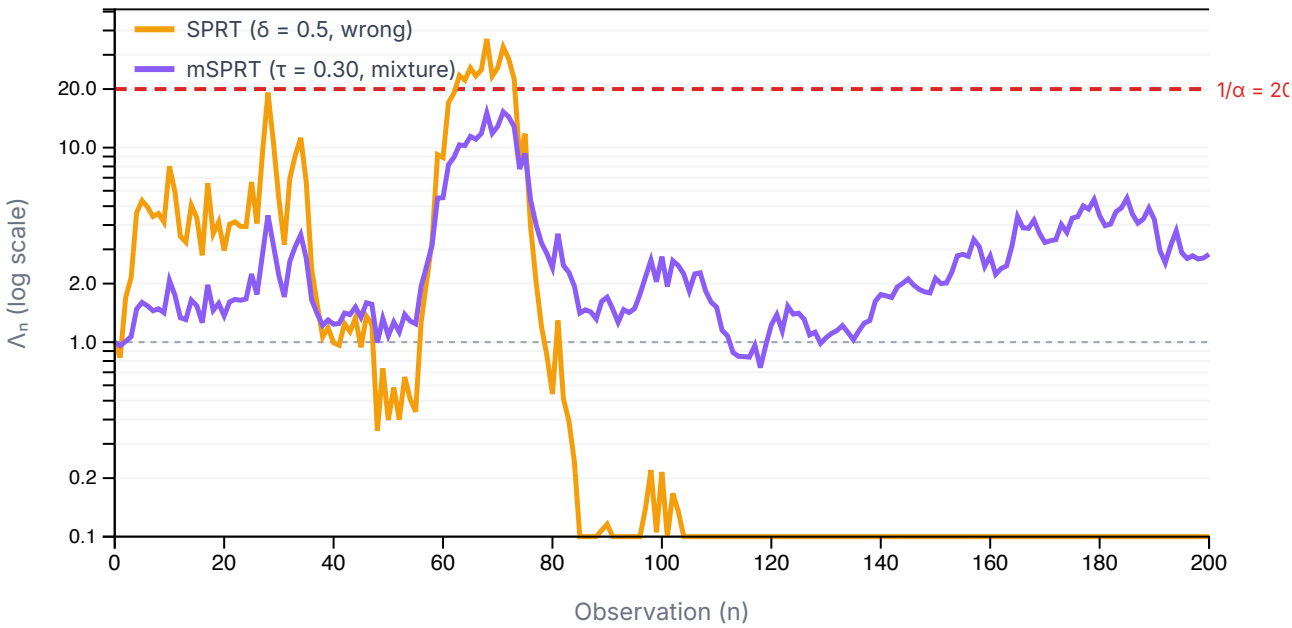
τ is therefore a *tuning knob*: pick it to match the smallest effect size you genuinely care about detecting. In Act 9 we will see that Eppo exposes this as the *target sample size* ν, which plays the same role under a different parameterisation.

SPRT vs mSPRT Comparison

SPRT assumes δ = 0.5 (wrong guess). mSPRT uses a mixture over effect sizes. The true effect is 0.20.

Run simulation

True effect: 0.20 Mixture τ: 0.30 Observations: 200



SPRT (wrong δ)
Reject at n=63

mSPRT (mixture)
Undecided

Key insight: When the true effect size differs from the SPRT's assumed δ, the SPRT can be slow or erratic. The mSPRT hedges across many effect sizes via the mixture, making it robust. Both are anytime-valid — the difference is power and speed.

Simulation takeaway: The mixture approach lets you test for a range of effect sizes, not just one. The anytime-valid guarantee is preserved.

Part B: The mSPRT for A/B Testing (Johari et al., 2017)

Johari, Pekelis, and Walsh (2017) applied Robbins' idea to the specific setting of A/B tests. They chose the Normal distribution as the mixing distribution H and derived a closed-form formula for the mixture likelihood ratio. They called it the **mixture**

Sequential Probability Ratio Test (mSPRT).

Their key contribution: making the theory practical. They gave explicit formulas, calibration guidance for τ, and proved that being within an order of magnitude of the true effect-size variance is sufficient for near-optimal power.

Closed form for Normal observations

Assumptions:

- Observations $x_i \sim \mathcal{N}(\mu, \sigma^2)$ (Normal with mean μ, **known** variance σ^2).
- Mixing distribution $H = \mathcal{N}(0, \tau^2\sigma^2)$ (Normal centred at 0, variance proportional to σ^2).

The mixture likelihood ratio (Johari et al., Proposition 1):

$$\Lambda_n^H = \frac{1}{\sqrt{1+n\tau^2}} \exp\left(\frac{n\tau^2\,\bar{x}_n^2}{2\,(\sigma^2/n)\,(1+n\tau^2)}\right)$$

Unpacking every piece:

Symbol	Meaning
σ^2	Variance of individual observations (how noisy the data is)
n	Number of observations so far
τ^2	Variance of the mixing distribution; controls how wide a range of effect sizes we consider plausible
$\bar{x}_n$	Sample mean after n observations
$1/\sqrt{1+n\tau^2}$	Shrinkage factor; slowly decreases as n grows
$\exp(\cdot)$	$e \approx 2.718$ raised to the argument; grows rapidly when $\bar{x}_n$ is far from zero

The mSPRT statistic is large when the sample mean is far from zero (strong signal) the sample size is large, and the observation noise is small relative to the mixing-

distribution variance.

Decision rule

$$\text{Stop and reject } H_0 \text{ when:} \quad \Lambda_n^H \geq \frac{1}{\alpha}$$

For $\alpha = 0.05$: stop when $\Lambda_n^H \geq 20$.

No lower boundary. Unlike the SPRT, the mSPRT cannot definitively confirm H_0. In practice, experiments run for a fixed maximum duration; if the threshold is never crossed, conclude: “No significant effect detected.”

Key assumption: known variance

The mSPRT formula assumes σ^2 is known. For continuous data this is typically estimated; for binary data (e.g. click/no-click), the exact mSPRT results hold only approximately via the Central Limit Theorem. For large samples ($n \geq 100$, nearly always true in A/B testing), the approximation is excellent.

Key Takeaway

Key concepts: Robbins' mixture idea, the mSPRT as a weighted average of likelihood ratios, the closed-form Normal formula, the role of τ, no lower boundary, the known-variance assumption.

What's next: The mSPRT was a landmark — but it is not what Eppo uses. The next act introduces **confidence sequences** fromHoward et al. (2021), a more general framework that Eppo adopted instead.

Act 9 — Confidence Sequences: The Modern Framework

A confidence sequence is the answer to the peeking problem for estimation. It gives you a range that is valid at every single time you check, not just one.

Key point: A standard 95% confidence interval is only valid at one pre-specified time. If you check repeatedly, the guarantee breaks. A confidence sequence is valid at all times, no matter how often you look.

The price: the band is wider than a fixed CI at any single time. But you are free to monitor continuously, stop whenever you want, and report at any time.

Simulation

Watch how a fixed 95% CI shrinks as more data arrives, but is invalid if you peek. The confidence sequence is wider at first, but always valid.

Confidence Sequence vs Fixed CI

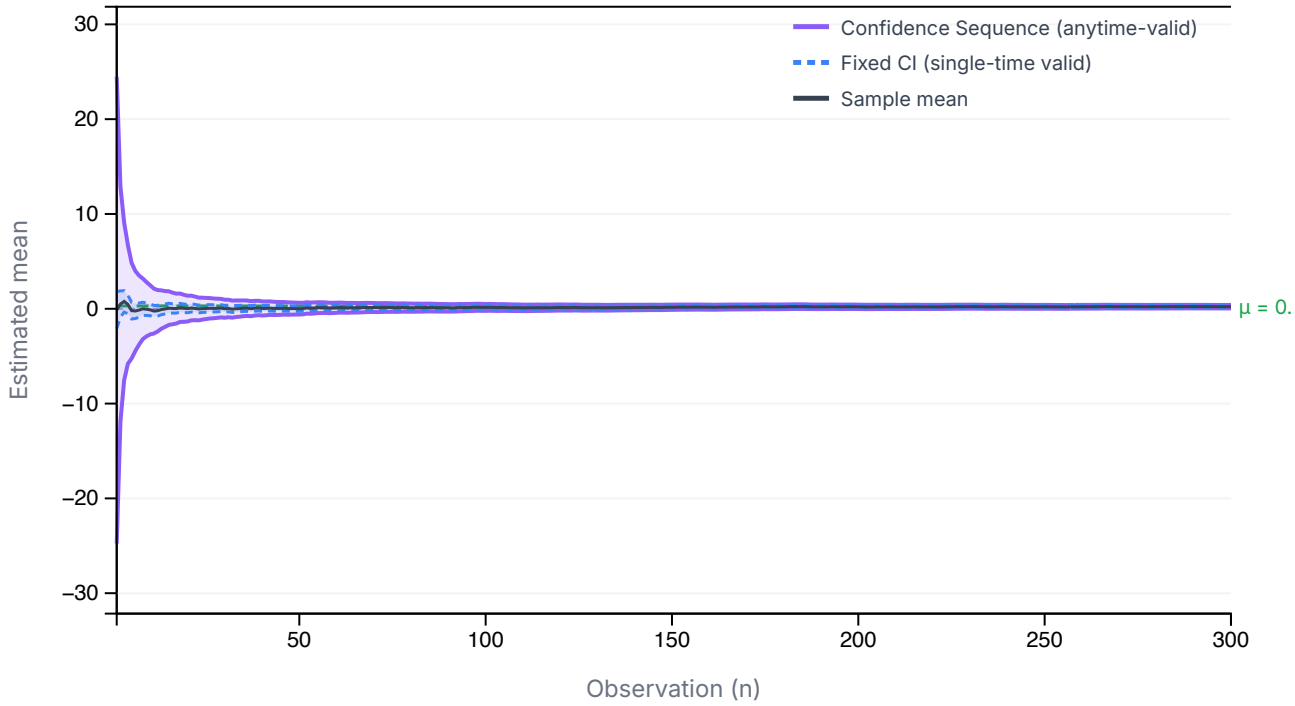
The CS (purple) is wider than the fixed CI (blue) — the “price of peeking” — but valid at all times simultaneously.

Run simulation

True effect μ: 0.30

Tuning v: 100

Observations: 300



0.226	0.362	1.60×
Fixed CI width at n=300	CS width at n=300	CS / CI ratio (price of peeking)

Simulation takeaway: Confidence sequences let you estimate with valid coverage at every time, no matter how often you peek. This is the modern solution for sequential estimation.

Mathematical Formulation

Definition

Definition (Confidence Sequence)

A sequence of intervals $(C_t)_{t\geq 1}$ is a $(1-\alpha)$-**confidence sequence** (CS) for a parameter μ if:

$$\mathbb{P}\big(\mu \in C_t \text{ for all } t \geq 1\big) \geq 1-\alpha$$

The **entire sequence** of intervals simultaneously covers the true parameter with

probability at least $1-\alpha$. Not just at one time — at every time.

The Normal Mixture Boundary (the key formula)

The centrepiece of Howard et al. is the **Normal mixture confidence sequence**. For sub-Gaussian observations with intrinsic time (cumulative variance) $v_t = \sum_{i=1}^{t} \sigma_i^2$:

$$u(v) = \sqrt{(v+\nu) \cdot \log \frac{v+\nu}{\nu\alpha^2}}$$

The $(1-\alpha)$-confidence sequence is:

$$C_t = \left(\bar{x}_t - \frac{u(v_t)}{t},\ \bar{x}_t + \frac{u(v_t)}{t} \right)$$

Unpacking the boundary:

Symbol	Meaning
v	**Intrinsic time**: the cumulative variance of observations so far. For i.i.d. data: $v_t = t\sigma^2$.
$\nu > 0$	**Tuning parameter**: controls which sample sizes the boundary is tightest for. The boundary is approximately optimised for $v \approx \nu$.
α	Significance level (e.g. 0.05)
$\log(\cdot)$	Natural logarithm

The boundary $u(v)$ tells you how far from zero the running sum S_t can drift before you can reject H_0. The parameter ν lets you tune where the boundary is tightest. If you expect to collect N observations total with variance σ^2, setting $\nu \approx N\sigma^2$ makes the boundary tightest around $n = N$.

Connection to Robbins and the mSPRT

Howard et al. explicitly credit Robbins (1970) as their starting point:

- **Robbins (1970):** introduced the mixture approach and confidence sequences, but only for known-variance Normal data.
- **Johari et al. (2017):** made it practical for A/B testing via the mSPRT, but still assumed known variance and parametric (Normal) data.
- **Howard et al. (2021): extended to nonparametric settings** via the sub-psi framework. The boundary works for *any* sub-Gaussian data (not just Normal), the variance can be unknown (estimated from data), and the coverage guarantee is exact (non-asymptotic).

Growth rate comparison

Method	Half-width rate
Fixed-sample CI	$\frac{\sigma}{\sqrt{n}} \cdot z_{\alpha/2}$
Normal mixture CS	$\frac{1}{n}\sqrt{(n\sigma^2+\nu)\log\frac{n\sigma^2+\nu}{\nu\alpha^2}}$
LIL-rate CS	$\frac{\sigma}{\sqrt{n}} \cdot \sqrt{2\log\log(n\sigma^2)}$

The Normal mixture CS is wider than the fixed CI by a factor of $\approx \sqrt{\log n}$ (the price of peeking). It is slightly wider than the LIL-rate CS asymptotically, but **tighter in practice**over the 2–3 orders of magnitude of sample sizes relevant to A/B testing.

Why Eppo chose this framework

1. **Anytime-valid:** Experimenters will peek — the guarantee must hold at every moment. ✓
2. **No known variance required:** Real experiments have unknown, potentially changing variance. ✓
3. **No distributional assumption:** Metrics like revenue or session length are far from Normal. ✓
4. **Tunable tightness:** Different experiments have different expected durations. ✓
5. **Produces a confidence interval:** Experimenters need to report “the lift is +3% ± 1.5%”, not just “significant / not significant.” ✓

Key Takeaway

Key concepts: confidence sequences (uniform coverage over all times), the Normal mixture boundary $u(v) = \sqrt{(v+\nu)\log\frac{v+\nu}{\nu\alpha^2}}$, the tuning parameter ν, the sub-ψ framework, nonparametric validity, the $\sqrt{\log n}$"price of peeking."

This is the framework Eppo uses. The boundary formula from this act is the engine inside Eppo's confidence interval.

Act 10 — The Problem Eppo Solves

Real A/B testing faces three problems at once:

1. **The peeking problem:** People check results often. Classical tests break under repeated checking.
2. **High variance:** User outcomes are extremely variable. Small effects are buried in noise, so experiments take a long time.
3. **Heterogeneous users:** Treatment effects differ across user segments. Pre-experiment data can help separate signal from noise.

Signal vs Noise

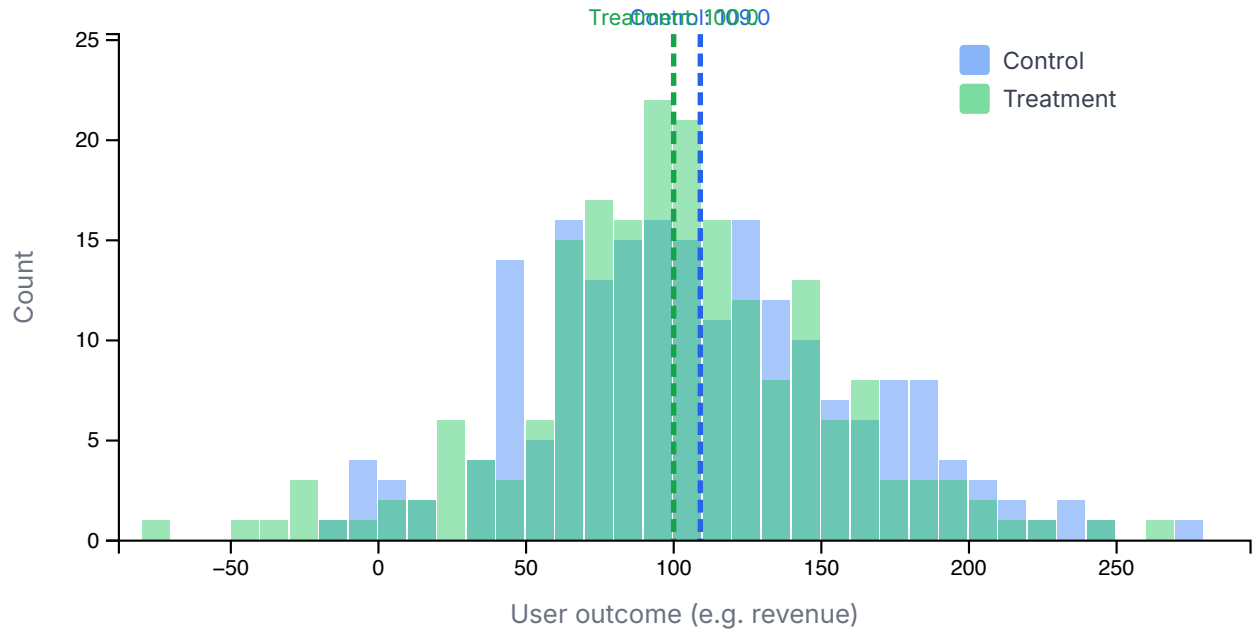
Blue = control, green = treatment. Try increasing noise — the distributions overlap and the signal disappears.

Run simulation

True effect: 2%

Noise (σ): 50

Users per group: 200



0.040	-9.03	±10.39	Not significant
Signal / Noise	Observed diff	95% CI half-width	p < 0.05?

The core problem: In real A/B tests, the true effect is tiny (e.g. +2%) while user-level noise is enormous (σ ≈ 200%). This is why experiments need thousands of users and why variance reduction (Act 11) is so valuable — it narrows the distributions, making the signal visible sooner.

What Eppo wants:

- **Anytime-valid inference:** Peek as often as you like; the error guarantee holds.
- **Faster experiments:** Use pre-experiment data to reduce noise, shrinking the confidence interval.
- **Simple outputs:** Report a confidence interval for the *relative lift* ("+3% ± 1.5%") that experimenters can act on immediately.

Acts 1–8 built each ingredient. The next three acts show how Eppo assembles them into one pipeline.

Key Takeaway

Three problems, three solutions:

Problem	Solution	Where we learned it
Peeking inflates errors	Sequential CI (Howard et al.)	Acts 6, 8
High variance / slow experiments	Regression adjustment (CUPED)	Act 12
Need interpretable output	Relative lift with CI	Act 13

Act 11 — The Eppo Pipeline, Step by Step

Eppo's pipeline takes raw data, removes predictable noise, and produces an anytime-valid confidence interval for the treatment effect. Each step builds on the previous acts.

`Randomise` → `Collect` → `Adjust` → `Estimate` → `CI` → `Decide`

Step 1: Randomise

Each user arriving at the website is randomly assigned to **control** (existing experience) or **treatment**(new feature). Randomisation ensures the groups are comparable — any difference in outcomes is caused by the treatment, not by pre-existing differences.

Step 2: Collect data over time

As users interact with the product, two quantities are recorded for each user:

- Y_i: the **outcome** during the experiment (e.g. purchase amount, clicks, session length).
- X_i: a **pre-experiment covariate** — the same metric from before the experiment started.

Step 3: Regression adjustment (noise removal)

This is where Eppo *generalises* the CUPED idea from Deng et al. (2013).

For each group **separately**:

1. Fit a regression model predicting Y from the pre-experiment covariates X.
2. Compute the **adjusted outcome** (residual) for each user:

$$Y_i^* = Y_i - \hat{f}_g(X_i)$$

Each adjusted outcome Y_i^*equals the raw outcome minus the model's prediction from pre-experiment covariates. The residual isolates the unpredictable component — the part that can be attributed to the treatment rather than to baseline user behaviour.

Step 4: Estimate the treatment effect

Compute the adjusted group means:

$$\hat{\mu}_0(t) = \frac{1}{n_0(t)} \sum_{i \in \text{control}} Y_i^*, \qquad \hat{\mu}_1(t) = \frac{1}{n_1(t)} \sum_{i \in \text{treatment}} Y_i^*$$

The estimated treatment effect and the **relative lift**:

$$\text{Relative lift}(t) = \frac{\hat{\tau}(t)}{\hat{\mu}_0(t)} = \frac{\hat{\mu}_1(t)}{\hat{\mu}_0(t)} - 1$$

If control averages €100 and treatment averages €103, the relative lift is $103/100$ -

$1 = 0.03 = +3\%.$

Step 5: Estimate the variance

Compute the standard error of $\hat{\tau}(t)$ from the **residuals** Y_i^*:

$$\hat{\sigma}^2_{\hat{\tau}}(t) = \frac{s_0^2(t)}{n_0(t)} + \frac{s_1^2(t)}{n_1(t)}$$

The variance is estimated from the data — not assumed known. This is a key difference from the mSPRT (Act 8), which required known variance. Howard et al.'s framework (Act 9) permits this.

Step 6: Construct the sequential confidence interval

Plug the estimated treatment effect and its standard error into the Normal mixture boundary from Act 9:

$$\mathrm{CI}(t) = \hat{\tau}(t) \pm \hat{\sigma}_{\hat{\tau}}(t) \cdot \sqrt{\frac{n+\nu}{n} \cdot \log \frac{n+\nu}{\nu\alpha^2}}$$

where $n = n_0(t) + n_1(t)$ is the total sample size and $\nu = M \cdot \hat{\sigma}^2$ is the tuning parameter.

Step 7: Decide

What you see	Colour	Action
CI entirely above 0	Green	Positive effect. Ship the feature.
CI entirely below 0	Dark grey	Negative effect. Revert.
CI crosses 0, experiment ongoing	Grey	Inconclusive. Keep collecting.
CI crosses 0, experiment ended	Grey	No significant effect. Decide pragmatic

Because the CI is a confidence sequence (Act 9), **this decision is valid at whatever time it is made.** No need to pre-specify an analysis time.

Try It: Interactive Pipeline

EPPO Pipeline Simulation

Watch the full Eppo pipeline in action: randomise users, collect data with CUPED adjustment, build sequential confidence intervals, and see when a decision is reached.

True effect size: 0.15

-0.3 (harm) 0 (null) 0.5 (strong)

Max sample size per group: 10.000

1,000 500,000 1,000,000

Run simulation

Key Takeaway

The Eppo pipeline in one sentence: Randomise → collect Y and covariates X → regression-adjust (remove predictable noise) → estimate lift and variance → wrap in a sequential CI → decide when CI excludes zero.

Act 12 — Variance Reduction via Regression Adjustment

User outcomes are noisy. If you can predict some of that noise using pre-experiment data, you can subtract it out and make your experiment much more efficient.

Key point: Regression adjustment removes predictable noise, leaving only the unpredictable part. This shrinks the confidence interval and speeds up decisions.

Variance Reduction: Raw vs CUPED

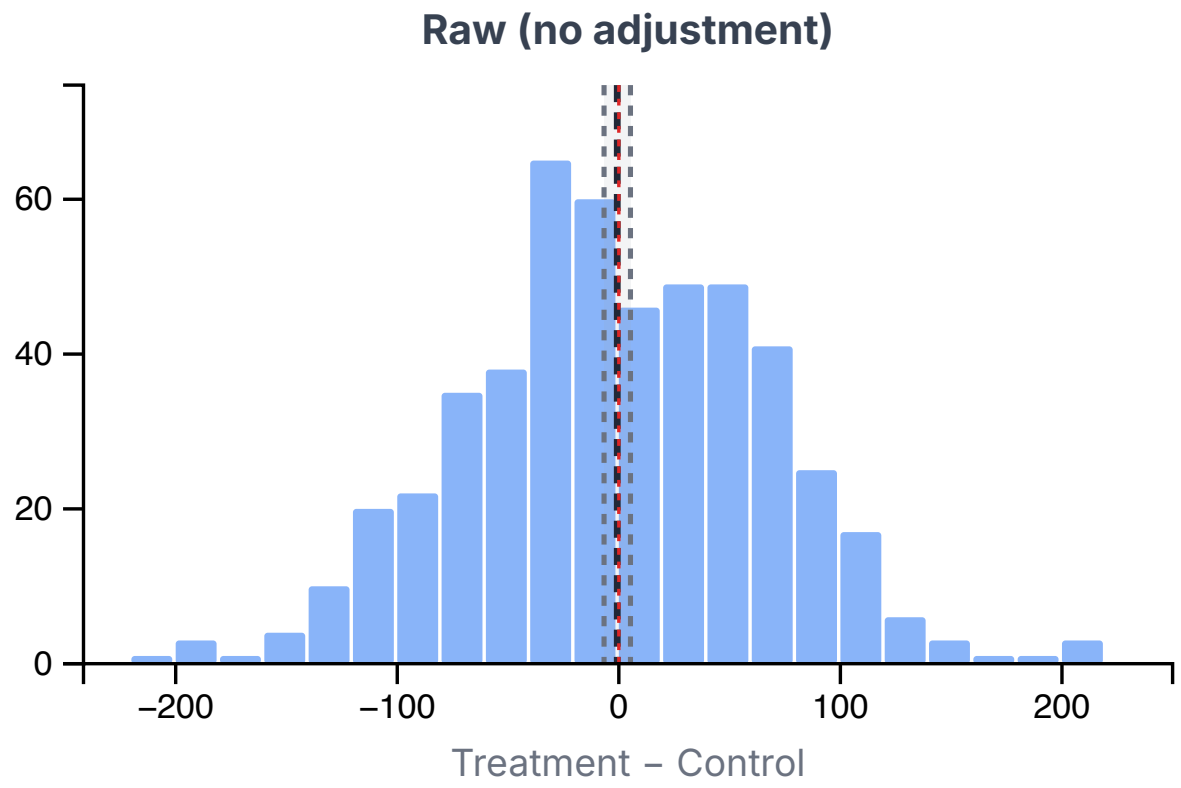
Left: raw differences. Right: CUPED-adjusted. Same data, tighter distribution. Dashed lines = 95% CI for the mean.

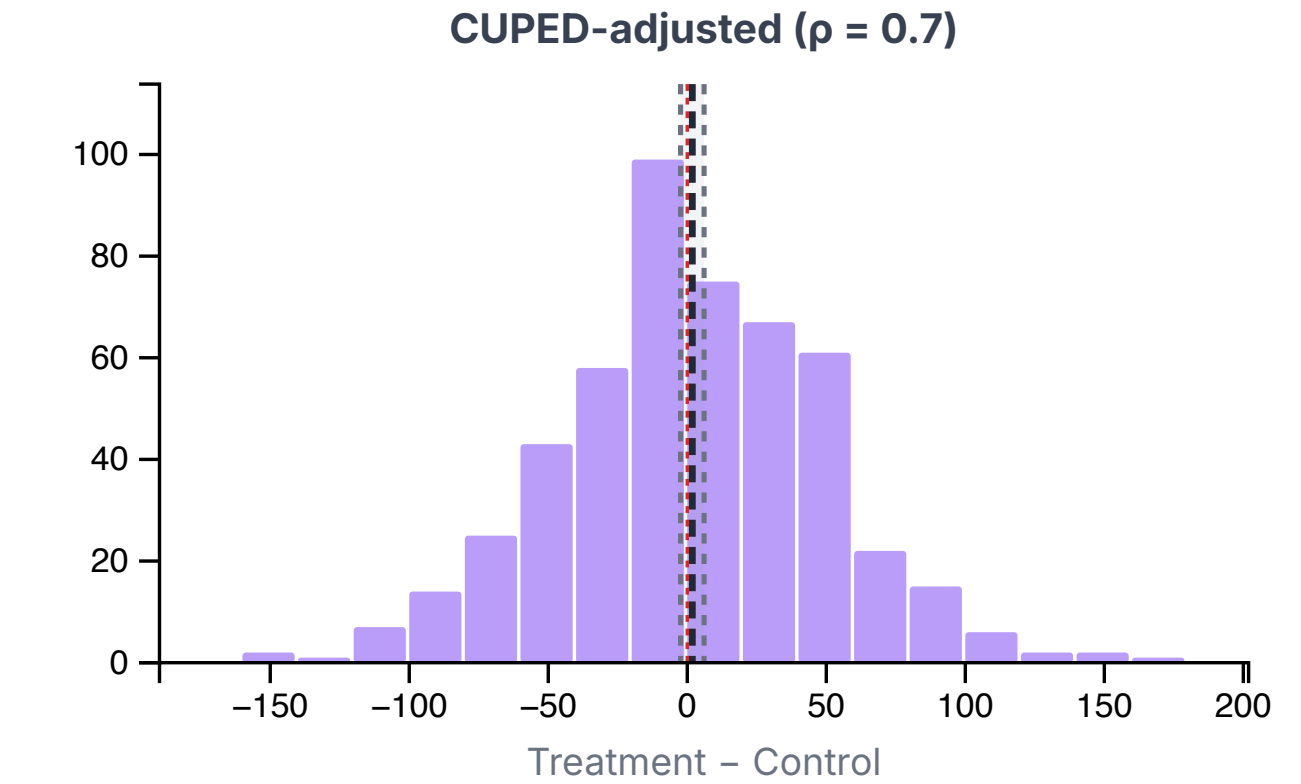
Run simulation

Correlation ρ: 0.70

True effect: 3%

Users per group: 500





±5.96	±4.23	49.7%	2.0×
Raw CI width	Adjusted CI width	Variance reduced (theory: 49.0%)	Effective sample multiplier

Mathematical Formulation

From CUPED to generalised regression adjustment

Recall CUPED (Deng et al., 2013):

$$Y^* = Y - \theta(X - \mathbb{E}[X])$$

with optimal $\theta = \mathrm{Cov}(Y, X)/\mathrm{Var}(X)$ and variance reduction $\mathrm{Var}(Y^*) = \mathrm{Var}(Y)(1 - \rho^2)$, where $\rho = \mathrm{Cor}(Y, X)$ is the correlation coefficient.

Eppo generalises this in two ways:

1. **Nonlinear models:** Instead of a single linear coefficient θ, fit a full regression model that can capture nonlinear relationships.
2. **Separate models per group:** Fit $\hat{f}_{\text{control}}(X)$ and $\hat{f}_{\text{treatment}}(X)$ independently.

$$Y_i^* = Y_i - \hat{f}_g(X_i)$$

The adjusted outcome Y_i^*equals the raw outcome minus the model's prediction from pre-experiment covariates. The residual captures only the unpredictable component of each user's behaviour.

Why this reduces variance

$$\mathrm{Var}(Y^*) = \mathrm{Var}(Y - \hat{f}(X)) = \mathrm{Var}(Y)(1 - R^2)$$

where $R^2$is the coefficient of determination — the fraction of variance in Y explained by X.

R^2	$1 - R^2$	Variance remaining	Traffic multiplier
0.10	0.90	90%	1.1×
0.25	0.75	75%	1.3×
0.50	0.50	50%	2×
0.60	0.40	40%	2.5×
0.75	0.25	25%	4×

With $R^2 = 0.50$, the regression explains half the variance. The remaining noise is halved — equivalent to running the experiment with twice as many users.

Critical Safety Rule

The covariate X must satisfy $\mathbb{E}[X^{(\text{treatment})}] = \mathbb{E}[X^{(\text{control})}]$. This is automatically guaranteed when X comes from the *pre-experiment* period (randomisation has not yet occurred).

NEVER use post-treatment covariates. Deng et al. demonstrated that using an in-experiment covariate produced **directionally opposite conclusions**, because the treatment itself changed X.

Why adjustment does not break the sequential guarantee

1. Covariates come from the pre-experiment period (independent of treatment).
2. Regression adjustment changes only the variance, not the mean treatment effect.
3. The sub-Gaussian condition (Act 9) is satisfied by the adjusted residuals.
4. Ville's inequality applies to the resulting supermartingale.

Regression adjustment reduces variance without changing the estimand. The sequential guarantee depends on the martingale structure, which is preserved because pre-experiment covariates are independent of treatment assignment.

Key Takeaway

Key concepts: regression adjustment as generalised CUPED, separate models per group, variance reduction factor $1 - R^2$, the safety rule (pre-experiment covariates only), adjustment preserves the sequential guarantee.

Act 13 — Sequential Confidence Intervals in Eppo

Eppo's confidence interval is valid at every time you check, not just one. This lets you monitor continuously and make decisions as soon as the evidence is clear.

Key point: The interval is a bit wider than a classical CI, but you can stop at any time

and the guarantee still holds.

The Formula, Revisited

From the pipeline (Act 11, Step 6):

$$\mathrm{CI}(t) = \hat{\tau}(t) \pm \hat{\sigma}_{\hat{\tau}}(t) \cdot \underbrace{\sqrt{\frac{n+\nu}{n} \cdot \log \frac{n+\nu}{\nu\alpha^2}}}_{\text{sequential multiplier}}$$

Compare with a standard CI:

$$\mathrm{CI}_{\text{classical}}(t) = \hat{\tau}(t) \pm \hat{\sigma}_{\hat{\tau}}(t) \cdot \underbrace{z_{\alpha/2}}_{\approx 1.96}$$

The only difference is the **sequential multiplier** replacing the constant 1.96.

How the sequential multiplier behaves

n	$m(n)$ **(example)**	$z_{\alpha/2}$	**Interpretation**
100	≈ 3.8	1.96	Early: sequential CI is ~2× wider
1,000	≈ 2.8	1.96	Mid-experiment: gap narrowing
10,000	≈ 2.3	1.96	Late: only ~17% wider
100,000	≈ 2.1	1.96	Approaching classical, never quite reaching it

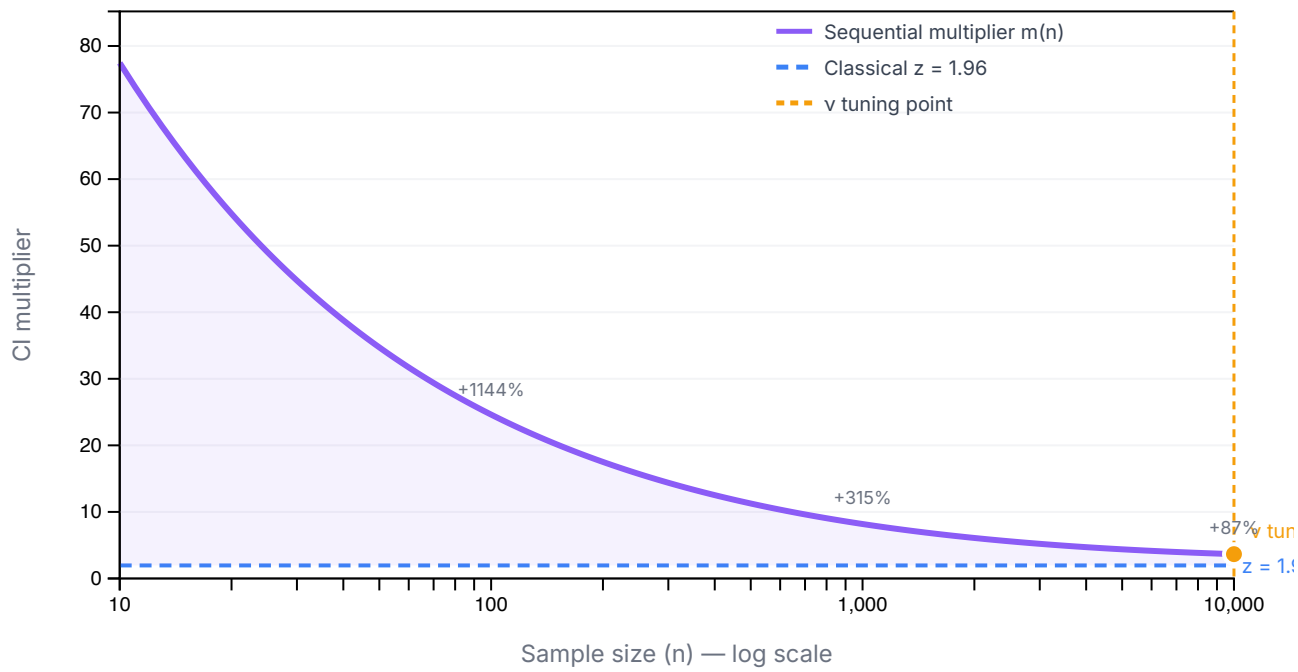
The sequential multiplier starts high and shrinks toward $z_{\alpha/2}$as more data arrives. The gap — the “price of peeking” — decreases but never fully closes. For most practical experiments (thousands to hundreds of thousands of observations), the price is modest: 10–40% wider than a classical CI.

Sequential Multiplier m(n)

The purple curve shows how the sequential CI multiplier approaches — but never reaches — the classical 1.96. The shaded area is the "price of peeking."

v tuning: 1.0× planned N → tightest at n = 10.000

Max sample size: 10.000



24.62	8.18	3.66
m(100) — 1156% wider than classical	m(1.000) — 317% wider than classical	m(10.000) — 87% wider than classical

The role of ν and M

Schmit & Miller set $\nu = M \cdot \hat{\sigma}^2$, where M is the **expected total sample size**.

If...	Then...
M is too small (underestimate traffic)	CI is tight early but wide late
M is too large (overestimate traffic)	CI is wide early but tight late
M matches actual traffic	CI is tightest around the planned end date

The parameter M controls where the boundary is tightest. Setting M equal to the planned total sample size optimises the CI width around the expected decision point.

What Eppo Actually Uses (Summary)

- **Framework:** Howard et al. (2021) confidence sequences (not the mSPRT of Johari et al. 2017).
- **Boundary:** The Normal mixture boundary $u(v) = \sqrt{(v+\nu)\log\frac{v+\nu}{\nu\alpha^2}}$.
- **Tuning:** $\nu = M\hat{\sigma}^2$, where M is the expected total sample size.
- **Variance:** Estimated from data (not assumed known).
- **Noise reduction:** Generalised CUPED via per-group regression adjustments.
- **Estimand:** Relative lift $\mu_1/\mu_0 - 1$.
- **Decision:** Stop when CI no longer crosses zero, valid at any time.

Act 14 — What If Your Experimentation Platform Does Not Support Sequential Testing?

Standard confidence intervals guarantee 95% coverage only at a fixed sample size.

Confidence intervals are only valid when the sample size is fixed in advance. If we monitor continuously and stop based on the data, that guarantee no longer applies.

By checking repeatedly and stopping when results look good, we give ourselves multiple chances to get a lucky fluctuation.

At a fixed sample size, the chance of a false positive is 5%. If we can stop at any time, the chance of ever making at least one false positive is much higher.

The issue isn't peeking itself — it's that standard statistical tools weren't designed for continuous monitoring.

The issue isn't peeking itself — it's that standard statistical tools weren't designed for continuous monitoring.

We need methods that remain valid no matter when we stop.

These methods control the probability of ever making at least one false positive, even if we keep checking.

Method 1: Bonferroni Correction (Simplest)

Before the experiment starts, commit to checking exactly K times. At each check, use a significance level of α/K instead of α.

$$\mathrm{CI}_k = \hat{\tau}(t_k) \pm \hat{\sigma}_{\hat{\tau}}(t_k) \cdot z_{\alpha/(2K)}$$

K (peeks)	α/K	$z_{\alpha/(2K)}$	CI multiplier vs. 1.96
1	0.050	1.96	1.00×
2	0.025	2.24	1.14×
4	0.0125	2.50	1.28×
12	0.00417	2.87	1.46×
52	0.00096	3.29	1.68× (weekly for a year)

Method 2: Pocock Boundaries (1977)

Like Bonferroni, pre-specify K equally-spaced analysis times. But instead of Bonferroni's crude α/K split, use a constant critical value c_P that accounts for the correlation between test statistics at different peeks.

K	c_P (Pocock)	$z_{\alpha/(2K)}$ (Bonf.)	Improvement

K	c_P (Pocock)	$z_{\alpha/(2K)}$ (Bonf.)	Improvement
2	2.18	2.24	3% tighter
4	2.36	2.50	6% tighter
12	2.60	2.87	9% tighter
20	2.69	3.02	11% tighter

Method 3: O'Brien–Fleming Boundaries (1979)

Instead of a constant threshold, use a threshold that *decreases* over time. At the first peek, demand very strong evidence; at the last peek, demand nearly the same evidence as a classical test.

$$c_k^{\mathrm{OBF}} \approx z_{\alpha/2} \cdot \sqrt{K/k}$$

Peek k (of $K = 4$)	c_k^{OBF}	c_P (Pocock)	$z_{\alpha/2}$ (classical)
1 (25% of data)	4.05	2.36	1.96
2 (50% of data)	2.86	2.36	1.96
3 (75% of data)	2.34	2.36	1.96
4 (100% of data)	2.02	2.36	1.96

O'Brien–Fleming is extremely conservative early (threshold 4.05 at the first peek) but barely penalises the final analysis (2.02 vs. 1.96 — only 3% wider). This is the most efficient approach: if you run the experiment to completion, you pay almost nothing for the option to peek.

Head-to-Head Comparison

	Bonferroni	Pocock	OBF	Eppo
Pre-specify K?	Yes	Yes	Yes	No
Threshold type	Constant	Constant	Decreasing	Continuous
Valid between peeks?	No	No	No	Yes
Variance reduction?	Manual	Manual	Manual	Built-in
Implementation	1 line	Lookup table	Formula	Platform

Which method should you use?

- No platform, minimal effort: Bonferroni. Commit to K peeks. Replace 1.96 with $z_{alpha/(2K)}$. Done.
- No platform, moderate effort: O'Brien–Fleming or Pocock. These are not overly conservative and are a good practical choice for most A/B tests.
- Need continuous monitoring or variance reduction: Use a platform like Eppo, but be aware that the sequential CI is often much more conservative than necessary for typical A/B testing.

The key point: even the simplest correction (Bonferroni) is far superior to uncorrected repeated testing. Any team that currently checks results without adjustment should adopt at least Bonferroni immediately.

Try It: Compare Methods

Head-to-Head Method Comparison

Compare how Bonferroni, Pocock, O'Brien–Fleming, and Confidence Sequences behave on the same data. All methods use 6 equally-spaced peeks.

True effect: 0.20

Sample size per group: 10.000

Run simulation

Historical Note

Stuart Pocock (1977) and **Peter O'Brien & Thomas Fleming** (1979) developed these group sequential methods for clinical trials, where stopping a trial early can save lives. Their work sits between Wald's SPRT (1945) and Howard et al.'s continuous framework (2021) in the historical timeline. The key insight of all three: sequential monitoring requires a wider critical region, but the magnitude of the adjustment can be minimised with the right boundary.

The Hybrid Approach — The Recommended Default

How to combine the best of fixed-horizon and sequential testing in practice.

Pure sequential testing buys you the right to peek and stop early. But it also pays a price — wider confidence intervals at any single look. For most teams, neither extreme ("fixed n" or "monitor every second") is the best default.

The **hybrid approach** uses sequential testing for *monitoring* and falls back on a

fixed-horizon analysis at a planned stopping date for the *final decision*. It gives you safe early stopping *and* avoids the pitfalls of stopping at an unrepresentative moment.

Why hybrid?

- **Weekday and seasonality effects.** Many products have day-of-week patterns. Stopping on a Thursday at 5pm because the sequential CI just excluded zero can produce decisions that don't generalise to a full weekly cycle. Forcing the final analysis to land on a planned multiple-of-7-days horizon protects against this.
- **Novelty / primacy effects.** User behaviour after a launch often differs from steady-state behaviour. A fixed minimum runtime guards against locking in an early, transient effect.
- **Power.** Sequential bands are wider than fixed-horizon CIs at the same α. If you almost always stop at the planned horizon anyway, the fixed CI gives you tighter intervals for the same data.

Recipe

Step	What you do	Why
1. Plan a horizon	Pick a target sample size ν covering at least one full weekly cycle (typically 1–4 weeks).	Avoids weekday bias; gives the sequential CI its tightest calibration point.
2. Monitor with the sequential CI	Watch the sequential confidence interval at any cadence you like.	Type I error stays at α regardless of how often you peek.
3. Allow safety stops	Stop the experiment early *only* if the sequential CI is on the wrong side of zero (clear harm) or extremely wide of the practical significance threshold.	Protects users without making early-stop the default.
4. Decide at the planned horizon	At $n = \nu$, run a single fixed-horizon analysis and use the standard CI to make the ship/no-ship decision.	Tightest possible interval; one clean look at a full cycle of data.

Step	What you do	Why
5. Long-running tests	For tests that must run for months (e.g. retention, low-traffic surfaces), drop the fixed step and rely on the sequential CI alone.	When the horizon is uncertain, sequential coverage is what you have.

Takeaway

Default to the hybrid: **sequential CI for monitoring, fixed CI at a planned horizon for decisions**. Use pure sequential only when the horizon is genuinely unknown.

Summary: The Through-Line

Act	Object	Represents	Key idea
0	Repeated p-values	The peeking problem	Checking repeatedly inflates FP
1	Random walk S_n	Position	SD grows as $\sqrt{n}$
2	Winnings M_n	Gambler's fortune	Martingale; Doob: $\mathbb{E}[M_\tau] = M_0$
3	Likelihood ratio Λ_n	Evidence against H_0	Comparing hypotheses flip by flip
4	Λ_n under H_0	LR is a martingale	The cancellation proof
5	Markov + Ville	Anytime-valid guarantee	$\mathbb{P}(\Lambda_n \text{ ever} \geq 1/\alpha) \leq \alpha$
6	SPRT	Wald's decision rule	Optimal but requires known δ
7	Λ_n^H (mixture LR)	Robust evidence	Robbins' mixtures → Johari's mSPR

Act	Object	Represents	Key idea
8	Confidence sequence	Modern framework	Howard: nonparametric, estimated σ^2
9	Peeking vs. variance	Three problems	Peeking, high variance, heterogeneous users
10	Eppo pipeline	Statistical pipeline	Randomise → adjust → CI → decide
11	Regression adjustment	Variance reduction	$Var(Y^*) = Var(Y)(1 - R^2)$
12	Eppo's sequential CI	Final output	Sequential multiplier; decide anytime
13	DIY alternatives	Simple approximations	Bonferroni, Pocock, OBF vs. Eppo

The core message: All fourteen acts build on the same mathematical structure — non-negative (super)martingales and Ville's inequality. The anytime-valid guarantee flows from a single elegant fact:

> *No stopping rule applied to a non-negative martingale can inflate the probability of exceeding a fixed threshold.*

The evolution across eight decades:

Ville (1939) → Wald (1945) → Robbins (1970) → Johari (2017) → Howard (2021) → Eppo (2024)

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Appendix

The sections below are optional deep dives. They cover variance reduction (CUPED) and the historical “A/B at platform scale” problem that motivated Eppo's pipeline. The main argument of the guide does not depend on them — come back when you want the full picture.

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